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ORIGINAL RESEARCH

MicroRNA-21 in Scleroderma Fibrosis and its Function in TGF-β- Regulated Fibrosis-Related Genes Expression

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Abstract Uncontrolled fibrosis in multiple organs is the main cause of death in systemic sclerosis (SSc), and transforming growth factor-β (TGF-β) activation plays a fundamental role in the process. Our previous study demonstrated that miR-21 was significantly up-regulated in SSc fibroblasts. Here, we found that TGF-β regulated the expression of miR-21 and fibrosis-related genes, and decreased Smad7 expression. Over-expression of miR-21 in fibroblasts decreased the levels of Smad7, whereas knockdown of miR-21 increased its expression. Further study using a reporter gene assay demonstrated Smad7 was a direct target of miR-21. Similar to human SSc, the expression of miR-21 increased in the bleomycin induced skin fibrosis. Inhibition of fibrosis by treatment with anti-fibrosis drug bortezomib restored the levels of miR-21 and Smad7. MiR-21 may function in an amplifying circuit to enhance TGF-β signaling events in SSc fibrosis, and suggesting that miR-21 may act as a potential therapeutic target.

Keywords Scleroderma · miR-21 · Smad7 · TGF-β

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Introduction

Scleroderma (systemic sclerosis; SSc) is a highly heterogeneous autoimmune disease with unknown etiopathogenesis progressive fibrotic replacement of normal tissue architecture in multiple organs which is a prominent pathological finding and distinguishing hallmark of SSc [1, 2]. Although the cellular and molecular mechanisms of SSc remain unclear, it is well documented that transforming growth factor β (TGF-β) plays a fundamental role in the process [3, 4]. TGF-β is a multifunctional cytokine, capable of regulating fibroblast proliferation, differentiation, apoptosis, inducing cytokine secretion, and, most importantly, up-regulating the synthesis of collagen and extracellular matrix [4]. TGF-β mediates fibrosis positively by activating its downstream mediators, Smad2 and Smad3, but negatively by its inhibitory factor Smad7. Smad7 is strongly and rapidly induced by TGF-β, associates with the activated TGF-β receptor, interferes with the activation of Smad2 and Smad3 by preventing their receptor interaction and phosphorylation, and negatively regulates TGF-β signaling [5]. Dong et al. reported a reduction of Smad7 expression in SSc-derived fibroblast cultures as compared with fibroblast cultures from unaffected areas of the same patients, suggesting that defective Smad7 feedback inhibition could play a role in TGF-β hyperresponsiveness in SSc [6] . The mechanisms of reduced expression of Smad7 in SSc have yet to be elucidated [7].

MicroRNAs(miRNAs) are short (~22 nucleotides), evolutionarily conserved, single-stranded non-coding RNAs, which bind to the 3'untranslated regions (3'UTRs) of target genes and thereby repress the translation of target genes and/or induce the degradation of target gene mRNA [8]. To date, it is thought that the estimated 1,000 miRNAs

encoded in the human genome are expressed in a tissue-specific and cell-type-specific manner. Individual miRNAs can target numerous mRNAs and individual mRNAs can be targeted by multiple miRNAs, providing vast regulatory potential [9]. MiRNAs are key regulators in tissue fibrosis. Many miRNAs are regulated by TGF-β, and have their predicted targets genes involved in matrix repair and remodeling, such as collagens, matrix metalloproteinase and integrins [10], as well as Smad signaling proteins. MiRNAs might cause the up-regulation or derepression of fibrotic genes and exhibit profibrosis or antifibrosis. Elevated expressions of profibrotic miRNAs or reduced expressions of antifibrotic miRNAs are likely to be important in the development of fibrosis in SSc [11, 12].

Our previous study using microarray and real-time PCR analysis demonstrated that miR-21 was significantly up-regulated in SSc skin samples and fibroblasts, which inversely correlated with the expression of Smad7; The bioinformatics analysis [(miRanda (http://microrna.sanger.ac.uk), TargetScan (http://www.targetscan.org), and PicTar (http://pictar.bio.nyu.edu)] indicated that the 3′UTR of Smad7 contained conserved potential seed sequences (AUAAGCUA) for miR-21[13]. Here, in an effort to identify the potential roles of miR-21 in SSc fibrosis, we showed that miR-21 was induced by TGF-β, and exerted profibrogenic effects through negatively regulated Smad7 in skin primary fibroblasts. These observations were further supported by the data obtained from bleomycin-induced skin fibrosis. Therefore, the up-regulation of miR-21 which targets Smad7 mRNA provided a mechanistic basis for fibrosis. Our data also suggested that strategies to decrease miR-21 expression may have therapeutic value in SSc.

Materials and Methods

Patients, Biopsy Specimens, and Cell Culture

Skin Biopsies and fibroblasts were from our previous study's samples, including 6 patients (4 diffuse and 2 limited cutaneous SSc patients) and 4 normal controls[13]. This study was approved by the Institutional Review Boards of Xiangya Hospital of Central South University, Changsha, Hunan, China. Informed consent was obtained from all patients. Primary skin fibroblasts were expanded by outgrowth culture at 37°C in a humidified atmosphere of 95 % air-5 % CO_2 in Dulbecco's modified Eagle's medium (DMEM; Gibco Invitrogen) supplemented with 10 % heat-inactivated fetal bovine serum (FBS, Hangzhou Sijiqing Biological Technology, China). Cells from passages 3–5 were used for the experiments. After culturing to an 80 %–90 % confluent condition, the cells were ready for the experiments.

After 24 h in the serum-reduced DMEM (0.5 % fetal calf serum), skin fibroblasts were stimulated for different time courses (0, 6, 12, 24 and 48 h) with different concentrations of TGF-β (0, 0.1, 1, 5 and 10 ng/ml) (R&D Systems).

Transfection Experiments

Primary skin fibroblasts (normal or SSc skin fibroblasts) from passages 3–5 were transfected with miR-21 mimics (5′-UAGCUUAUCAGACUGAUGUUGA-3′) or miR-21 inhibitors (5′-UAGCUUAUCAGACUGAUGUUGA-3′), or with scrambled controls (miR-21 mimics/inhibitors/scrambled controls; QIAGEN) at a final concentration of 20 μM with the use of Hiperfect reagent (QIAGEN). At 48 h after transfection, cellular lysates were collected to analyze the expression of fibrosis-related genes: Collagen type I, alpha 1 and 2 (COL1A1, COL1A2), α-smooth muscle actin (α-SMA), and fibronectin (Fn). Additionally, normal skin fibroblasts were treated with TGF-β(10 ng/ml) after transfected with miR-21 mimics or inhibitors or scrambled controls, and lysed after 48 h to measure fibrosis-related genes.

Histological Analysis and miRNA In Situ Hybridization

The skin biopsy specimens were fixed with 4 % paraformaldehyde solution and embedded in paraffin. Some skin sections (5 μm thick) were prepared and stained with hematoxylin and eosin for histological examination (H&E). In situ hybridizations (ISH) were performed overnight at 53 °C after the addition of 50 nM digoxigenin (Dig)-labeled locked nucleic acid (LNA)-based probe specific for miR-21 (Exiqon, Vedbaek, Denmark). Next, washing and scanning were carried out according to the manufacturer's protocols. The intensity of ISH stain from tissues and fibroblasts was calculated by Image Pro Plus Software 6.0 (Media Cybernetics, CA, USA). The mean density of cells was quantified and used for statistics analysis.

RNA Isolation and Quantitative Real-Time PCR Analysis

Total RNA from the skin biopsy specimens and fibroblasts were isolated with the miRNeasy mini kit (Qiagen). ①The miScript reverse transcription kit (Qiagen) and miScript SYBR Green PCR kit (Qiagen) were used to measure the expression levels of miR-21 in a model 7500 real-time PCR system analyzer (Applied Biosystems), with a miR-21 specific primer and the miScript Universal Primer (Qiagen). Expression of the U6B small nuclear RNA (RNU6B) was used as the endogenous control to normalize the sample data. ②To determine the quantity of mRNA, cDNA was prepared by Reverse Transcription System (Promega, USA) and amplified by real-time PCR with SYBR Green (SYBR Premix Ex Taq RT-PCR kit; Takara, Shiga, Japan) with the

appropriate primers (listed below). Expression of the GAPDH was used as the endogenous control to normalize the sample data. Relative expression levels were calculated with the $2^{-\Delta\Delta Ct}$ method[14]. The following primers were used: human GAPDH forward: 5′-GCACCGTCAAGGCTGAGAAC-3′, GAPDH reverse: 5′-TGGTGAAGACGCCAGTGGA-3′; human Smad7 forward:

5′-TGCTGTGCAAAGTGTTCAGGTG-3′, Smad7 reverse: 5′-CCATCGGGTATCTGGAGT AAGGA-3′; human COL1A1 forward:

5′-TCTAGACATGTTCAGCTTTGTGGAC-3′, COL1A1 reverse: 5′-TCTGTACGCAGGTGATTGGTG-3′; human COL1A2 forward: 5′-GAGGGCAACAGCAG GTTCACTTA-3′, COL1A2 reverse: 5′-TCAGCACCACCGATGTCCA-3′; human α-SMA forward: 5′-GACAATGGCTCTGGGCTCTGTAA-3′, α-SMA reverse: 5′-CTGTGCTTCGTCACCCAC GTA-3′; human fibronectin forward: 5′-ACCCA GTCCCGAAGGCACTA-3′, fibronectin reverse: 5′-CCTGTACATCTAAAGGCATGAAGCA-3′. Mouse GAPDH forward: 5′-TGTGTCCGTC GTGGATCT GA-3′, GAPDH reverse: 5′-TTGCTGTTGAAGTCG CAGGAG-3′; mouse Smad7 forward: 5′-CTGC TGTGCAAAGTGTTCAGG-3′, Smad7 reverse: 5′-CCATTGGGTATCTGGAGTAAGGA-3′; mouse COL1A1 forward: 5′-GACATGTTCAGCTT TGTGGACCTC-3′, COL1A1 reverse: 5′-GGGACC CTTAGGCCATTGTGTA-3′; mouse COL1A2 forward: 5′-TGCTTGCAGTAACTTCGTGCCTA-3′, COL1A2 reverse: 5′-CATGGGACCATCAACACCATC-3′; mouse α-SMA forward: 5′-AAGAGCATCCGACACTGCTGAC-3′, α-SMA reverse: 5′-AGCACAGCCTGAATAGCCACATAC-3′; mouse fibronectin forward: 5′-GCTTTGGCAG TGGTCATTTCAG-3′, fibronectin reverse: 5′-ATTCCCGAGGCATGTGCAG-3′.

Western Blot Analysis

Samples containing equal amounts of protein were subjected to 12 % SDS-PAGE and transferred to a polyvinylidene difluoride membrane (0.2 μm; GE Healthcare, USA). After being blocked with 2 % bovine serum albumin, the blots were probed with anti-Smad7 antibody (Abcam), and anti-GAPDH antibody (Cell Signaling Technology). After being washed with tris-buffered saline with Tween 20, the membranes were treated with horseradish peroxidase-conjugated secondary antibody at room temperature for 1 h and visualized by enhanced chemiluminescence (Amersham Pharmacia Biotech).

Plasmids and Constructs

The full-length 3′UTR (1,491 bp) of human Smad7 gene was amplified by PCR using human cDNA as a template. The primers were: sense 5′-TCGGAGCTCGAGGTCAT CTTCAACAGCCG-3′ and antisense 5′-GCGAAG CTTACATTCAGCTAGGTGATA AC-3. The PCR fragment was cloned downstream of the luciferase gene between the SacI and HindIII sites in pMIR-Report (Applied Biosystems). To generate Smad7 3′UTR substitution mutant (Smad7 3′ UTRS) containing mutations in the conserved miR-21 binding site, site-directed mutagenesis was performed using the wild-type 3′UTR as the template. In the 3′UTRS mutant, the nucleotide sequence complementary to nt 2–5 of miR-21 was mutated to the same sequence as that in miR-21 (from AGCU to TCGA). To generate Smad7 3′UTR deletion mutant (Smad7 3′ UTRD) containing a deletion of the conserved miR-21 binding site, site-directed deletion was performed using the wild-type 3′UTR as the template.

Luciferase Reporter Assay for Targeting the 3′ UTR of Smad7

Transient transfections were performed using Lipofectamine 2000 reagent (Invitrogen) according to the manufacturer's instructions. Briefly, cells were seeded at a density of 4×10^4 cells/well in a 24-well plate for 24 h before transfection. For each well of cells, 0.8 μg of the luciferase reporters containing the wild-type or a substitution mutation (Smad7 3′ UTRS) or a deletion mutation (Smad7 3′ UTRD) were cotransfected with miRNA scramble or miR-21 mimics (0.075 μg) into HEK 293 cells. As an internal control, renilla luciferase reporters (0.02 ug) were also included. 48 h after transfection, the cells were collected and dual luciferase activities were measured according to the manufacturer' s instructions (Promega).

Bleomycin-Induced Skin Fibrosis in Mice

The animal protocol used in this study was approved by the local animal care and use committee. Skin fibrosis was induced in 6-week-old female DBA/2 mice by local intracutaneous injections as described previously [15, 16] . The mice were challenged with intracutaneous injections of bleomycin (100 μl of a 0.5 mg/ml solution in 0.9 % NaCl) in defined areas of 1.5 cm^2 on the upper back every other day for 6 weeks.

One group of mice was treated with intraperitoneal injections of anti-fibrosis drug bortezomib (400 μg/kg twice weekly for the final 3 weeks of the experiment) to assess the impact of bortezomib on miR-21 levels in established fibrosis [17]. Another group was treated with

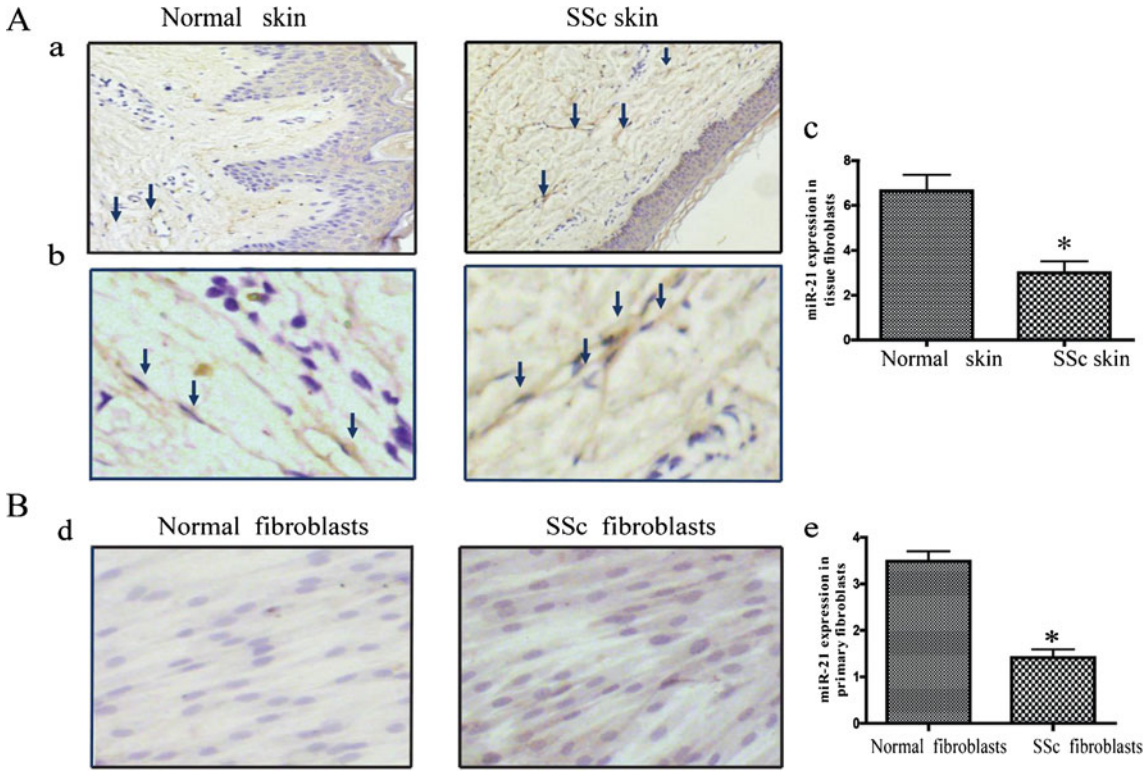


Fig. 1 Expression and localization of miR-21 in normal and SSc skin tissues. Skin tissues were obtained from 4 normal controls and 6 SSc patients, the expression and localization of miR-21 in skin tissues (**A**) and fibroblasts (**B**) were determined by in situ hybridization (ISH). $^{*}P$ <0.05 compared with the normal controls. Values are means ± SEM. The magnification of Figure a is 4x, Figure b, d is 10x

0.9 % NaCl instead of bortezomib for the final 3 weeks. The control group received intracutaneous injections of 100 μl 0.9 % NaCl for 6 weeks instead of bleomycin and additionally received 0.9 % NaCl for the final 3 weeks. After 6 weeks, animals were asphyxiated with CO_2. The upper back skin was removed and processed for further analysis. Dermal thickness was analyzed with a Olympus IX71 microscope (Olympus IX71, Olympus Optical Co. Ltd, Tokyo, Japan) by measuring the maximal distance between the epidermal–dermal junction and the dermal–subcutaneous fat junction at 4 different skin sections in each mouse. Hypodermal thickness was determined by measuring the thickness of the subcutaneous connective tissue beneath the panniculus carnosus at 4 different sites of the upper back in each mouse [15].

Statistical Analysis

GraphPad Prism software was used for statistical analysis. Numerical variables with normal distribution were compared with unpaired *t*-test or paired *t*-test. Non-normal distribution data were compared with Wilcoxon rank sum test. Data were expressed as mean ± SEM. $P<0.05$ was considered statistically significant.

Results

MiR-21 Expression was Up-regulated in SSc

In previous study, we identified that the expression of miR-21 was up-regulated in SSc skins and fibroblasts by real-time PCR. In order to futher identify the function of miR-21 in the pathogenesis and progression of SSc fibrosis, we used in situ hybridization (ISH) with an LNA miR-21-specific probe to examine the expression and the subcellular localization of miR-21 in human skin tissues and primary fibroblasts. ISH of skin tissues and primary fibroblasts showed cytoplasmic staining for miR-21, and the expression of miR-21 in SSc tissues and primary fibroblasts was significantly higher than normal controls (Fig. 1 A, B).

MiR-21 Regulated the Expression of Fibrosis-Related Genes

TGF-β is a central pathological mediator of SSc fibrosis, it plays a dominant role in the production and deposition of collagens and induces a transition of fibroblasts into myofibroblasts. MiR-21 expression was

increased in normal skin fibroblasts exposed to TGF-β, and the expression of miR-21 induced by TGF-β was in a dose- and time-dependent manner (Fig. 2 A, D). Consistent with bioinformatics prediction, up-regulation of miR-21 correlated with decreased expression of Smad7 in fibroblast (Fig. 2 B, E). Because heightened TGF-β has been associated with the promotion of fibrosis and extra cellular matrix deposition, we analysized the effect of TGF-β on fibrosis-related genes, COL1A1, COL1A2, α-SMA and Fn expression were increased in a dose- and time-dependent manner in fibroblasts (Fig. 2 C, F). The protein level of Smad7 was decreased at 48 h after TGF-β treatment (Fig. 2 H).

Overexpression and Knockdown of miR-21 Influenced the Expression of Fibrosis-Related Genes in Normal or SSc Fibroblasts

We next explored the potential functional effects of altered miR-21 expression on fibrogenesis. As we have shown that miR-21 was a central player in SSc fibrosis, we expected that up-regulation of miR-21 in normal skin fibroblasts may induce an SSc-like phenotype, with an increase in the expression of fibrosis-related genes, whereas restoration of miR-21 levels would decrease fibrosis levels in SSc fibroblasts. In support of our notion, increasing miR-21 levels by transfection with miR-21 mimics in fibroblasts were sufficient to induce the up-regulation of fibrosis-related genes, whereas decreasing miR-21 by siRNA attenuated the expression of these genes.

In normal fibroblast, overexpression of miR-21 was sufficient to cause the up-regulation of fibrosis-related genes (COL1A1, COL1A2, α-SMA, Fn) expression, and more distinct after stimulation with TGF-β (Fig. 3 A-a, b, c, d, e); Knocking down miR-21 attenuated the expression of fibrosis-related genes (COL1A1, COL1A2, α-SMA, Fn), and also more distinct after stimulation with TGF-β (Fig. 3 A-a, b, c, d, e); The increase or decreased of miR-21 expression was associated with a decrease or increased of Smad7 on the protein level (Fig. 3 A-f).

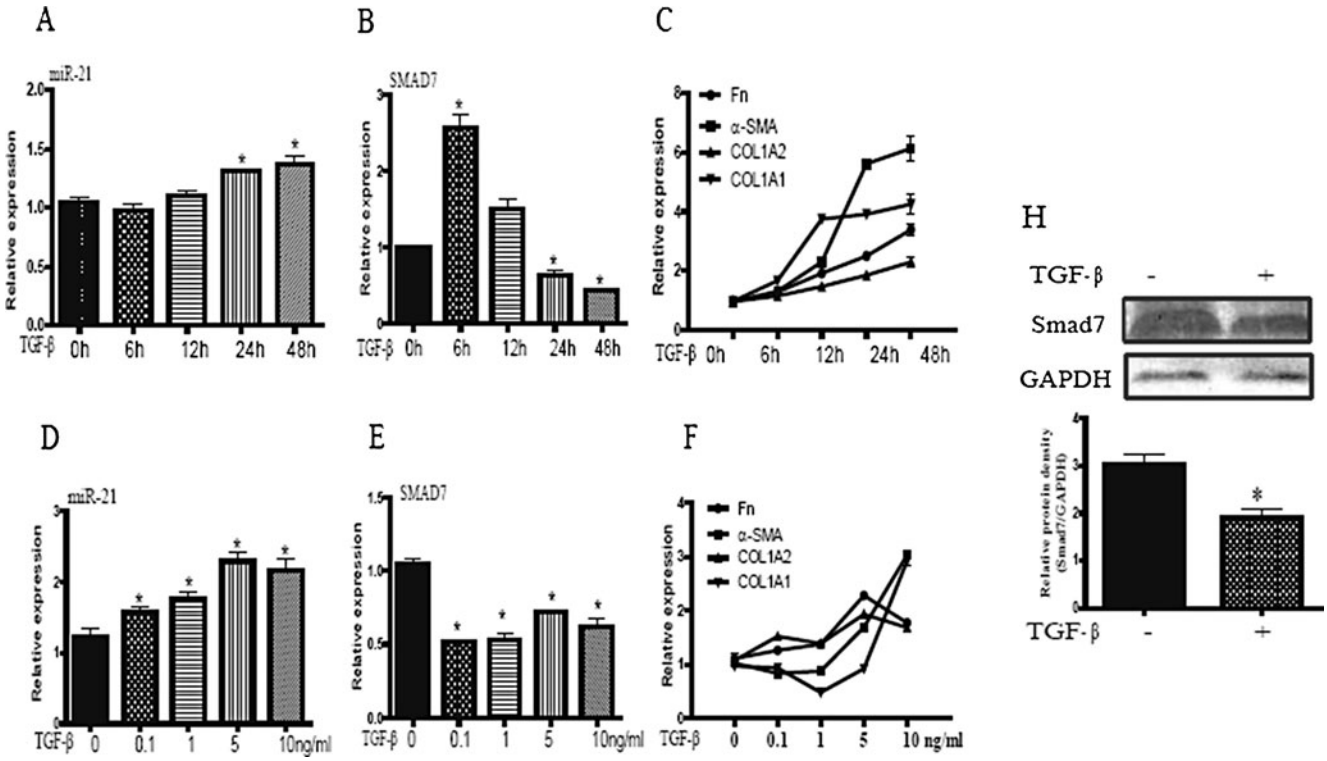
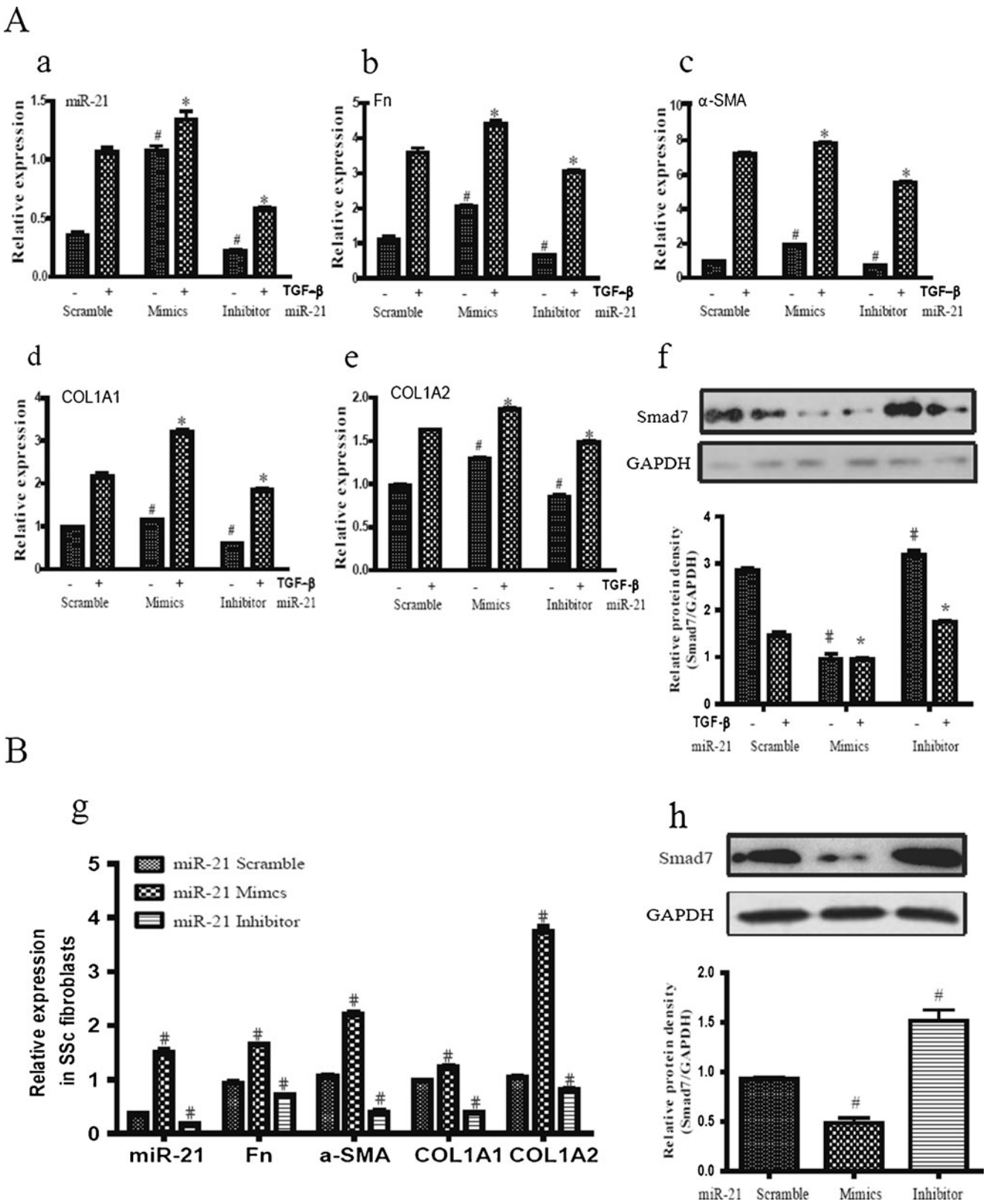


Fig. 2 The expression of miR-21 and fibrosis-related genes was regulated by TGF-β. Real-time PCR of cDNA prepared from fibroblasts exposed to TGF-β over different time periods (0, 6, 12, 24 and 48 h) or different concentrations (0, 0.1, 1, 5 and 10 ng/ml) was used to quantify RNA. MiR-21 is induced by TGF-β (**A** and **D**), expression pattern of Smad7 were shown (**B** and **E**), mRNA levels of fibrosis-related genes Fn, a-SMA, COL1A1, COL1A2 were determined (**C** and **F**). Cells were stimulated with TGF-β during 48 h and then harvested, and protein levels of Smad7 and GAPDH were determined (**H**). $^{*}P$ <0.05 compared with the TGF-β treated 0 h fibroblasts. Experiments were repeated at least 3 times. Values are means ± SEM



In SSc fibroblast, overexpression of miR-21 up-regulated the expression of fibrosis-related genes mRNA (Fig. 3 B-g); Knockdown of miR-21 decreased the expression of fibrosis-related genes mRNA (Fig. 3 B-g); Enhanced expression of miR-21 decreased whereas reduced expression of miR-21 increased the protein level of Smad7 (Fig. 3 B-g). Date

◀ **Fig. 3** Expression of fibrosis-related genes was altered following overexpression and knockdown of miR-21. **A** Normal human skin Fibroblasts were transfected with miR-21 mimics, miR-21 inhibitors and scramble as control; 48 h after transfection, the fibroblasts were treated with 10 ng/ml TGF-β for 48 h. MiR-21 levels(a) and mRNA levels of Fn (b), a-SMA (c), COL1A1 (d) and COL1A2 (e) were determined by real-time PCR, Smad7 protein levels (f) were determined by western blot. **B** SSc skin Fibroblasts were transfected with miR-21 mimics, miR-21 inhibitors and scramble as control; MiR-21 levels and mRNA levels of Fn , a-SMA, COL1A1 and COL1A2 were determined by real-time PCR (g), Smad7 protein levels (h) were determined by western blot. $^{\#}P<0.05$ compared with scramble mir-21 transfected fibroblasts without TGF-β stimulation. $^{*}P<0.05$ compared with scramble mir-21 transfected fibroblasts with TGF-β stimulation. Experiments were repeated at least 3 times. Values are means ± SEM

showing the effects of overexpression and knockdown of miR-21 on fibrosis may be mediated by Smad7.

Smad7 Was a Direct Target of miR-21

To determine whether Smad7 is the direct target for miR-21, we generated a luciferase reporter gene system by cloning full length of the 3'UTR of Smad7 (1,491 bp) with binding sites for miR-21 into the pMIR reporter vector. Analogous clones were prepared for the 3'UTR mutant that contains a substitution mutation (Smad7 3′ UTRS) and the deletion mutation (Smad7 3′ UTRD) in the predicted miR-21 seeding sequence (Fig. 4a). In the reporter gene assay, cotransfection of HEK 293 cells with miR-21 mimic and the luciferase reporter plasmid containing the 3′UTR of Smad7 decreased the relative luciferase activity by compared with cells transfected with the scrambled controls and plasmid (Fig. 4b), whereas cotransfection with miR-21 mimic and the luciferase reporter

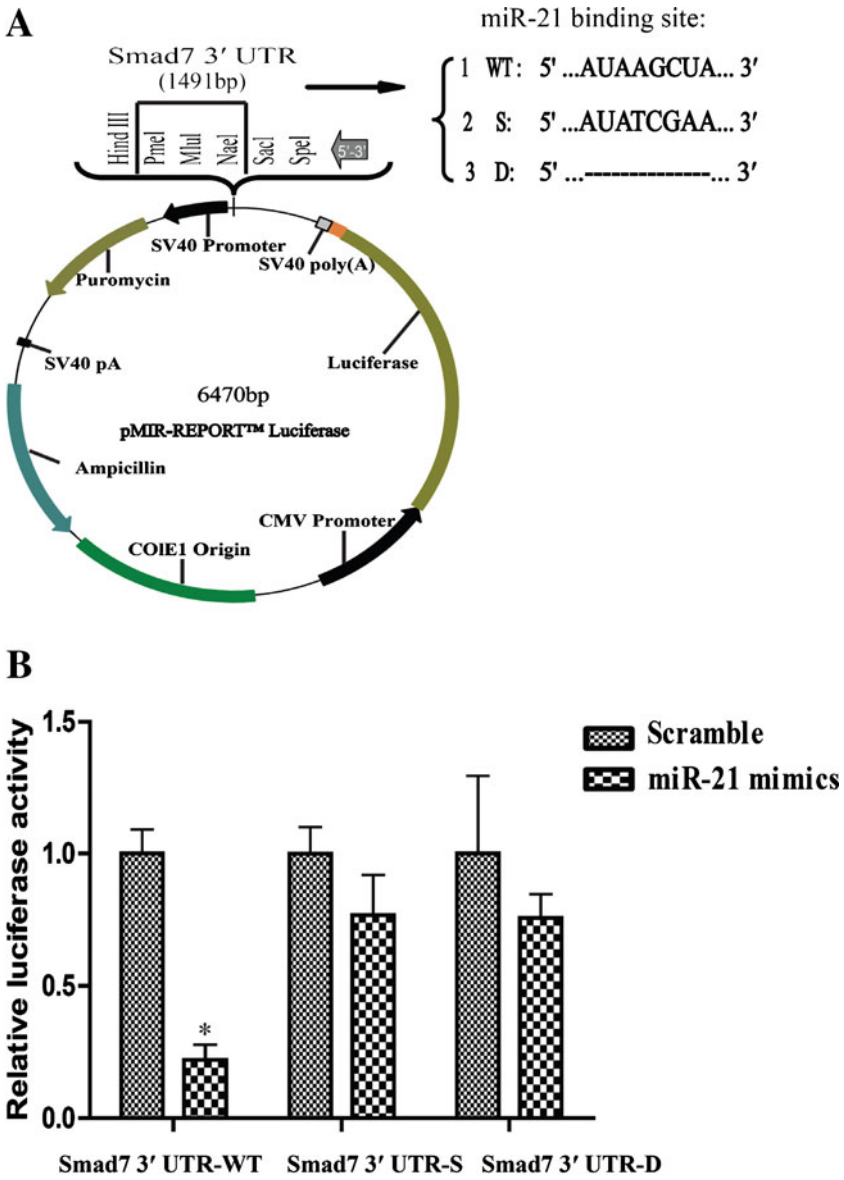


Fig. 4 Smad7 was directly regulated by miR-21. **a** The construction of luciferase reporter gene system: wile-type (WT) contains the full length of the 3'UTR of Smad7 (1,491 bp) with binding sites for miR-21, the Smad7 3′ UTRS contains a substitution mutation and the Smad7 3′ UTRD contains the deletion mutation in the predicted miR-21 seeding sequence. **b** Increasing the amounts of the miR-21 by transfection of miR-21 mimics represses the luciferase activity of Smad7 3′ UTR-wt, but has no influence on the mutants (Smad7 3′ UTRS or Smad7 3′ UTRD). $^{*}P<0.05$ compared with transfection scramble

plasmid containing the Smad7 3′ UTRS and Smad7 3′ UTRD did not show significant effects (Fig. 4b). In summary, The Smad7-expressing vector containing wild-type, but not mutant Smad7 3′UTR, was subject to regulation by miR-21 (Fig. 4b). This indicated Smad7 was directly regulated by miR-21.

MiR-21 Acted in the Murine Model of Bleomycin-Induced Skin Fibrosis

Finally, we wanted to investigate whether the increased levels of miR-21 in human SSc are mimicked in the SSc model of bleomycin-induced skin fibrosis in mice. Similar to the findings in human SSc, compared with normal controls (i.e., mice treated with saline instead of bleomycin for 6 weeks), administration of bleomycin increased dermal thickness and fibrosis-related genes expressions, whereas treatment with potential anti-fibrotic drug bortezomib for the final 3 weeks of the experiment not only stopped further progression of fibrosis but also decreased dermal thickness and reduced the expression of fibrosis-related genes and Smad7 (Fig. 5 A, B).

Therefore, we demonstrated that the increased levels of miR-21 in human SSc can be mimicked in the SSc model of bleomycin-induced skin fibrosis in mice. Further supporting the notion that miR-21 was a central player in SSc, and that miR-21 targeted Smad7 for down-regulation which eventually leaded to heightened TGF-β signaling, thus eliciting fibrosis.

Discussion

Fibrosis in multiple organs is a prominent pathological finding and distinguishing hallmark of SSc. There is no therapy has been shown to reverse or arrest the progress of fibrosis, which reflects the incomplete understanding of the molecular pathogenesis of this disease[1]. miRNAs recently emerged as ritical players in many pathophysiological processes and have been shown to participate in the progression tissue development, differention, repair and fibrosis[10, 18,

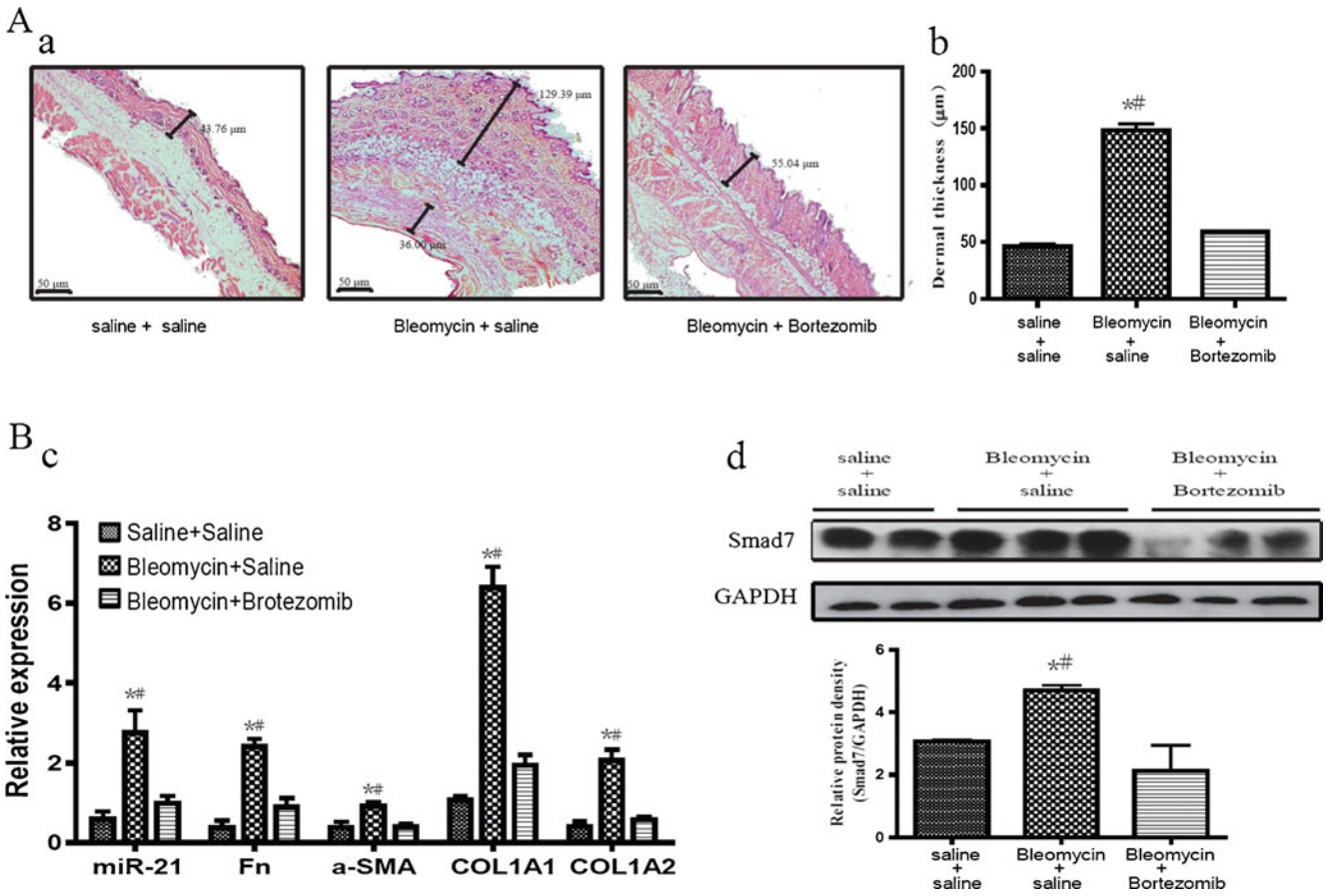


Fig. 5 MiR-21 acted in the murine model of bleomycin-induced skin fibrosis. (**A**) Histopathology of skin lesions in bleomycin-induced mice characterized by an excessive deposition of collagens and deep fibrosis, skin fibrosis was ameliorated in bortezomib-treated group. Real-time PCR was used to analyze the expression of miR-21 and fibrosis-related genes (**B**-c) in normal controls, bleomycin-induced mice and bortezomib-treated mice. The expression of Smad7 protein were also determined (**B**-d). *$P<0.05$ compared with normal control group, #$P<0.05$ compared with bortezomib-treated group. Experiments were repeated at least 3 times. Values are means ± SEM. Each group $n=10$

19]. However, how miRNAs regulate SSc fibrosis remains unclear. Elevated expression of profibrotic miRNAs or reduced expression of antifibrotic miRNAs are likely to be important in the development of fibrosis in SSc [1].

In this study, we examined the role of miR-21 in fibrosis using SSc skin bioipsy samples and bleomycin-treated mice. We found the expression of miR-21 was up-rgualated in SSc fibrosis and bleomycin-treated mice. MiR-21 was induced by TGF-β and in turn promoted TGF-β-regulated fibrogenic activation of skin fibroblasts by targeting Smad7. MiR-21 appears to function in an amplifying circuit to enhance TGF-β signaling events and to promote SSc fibrosis. Previous studies showed that miR-21 regulates lung, heart, renal fibrosis by enhancing the fibrogenic activity or promoting the proliferation of interstitial fibroblasts [20–22]. MiR-21 participates in fibrogenic events by regulating a unique array of targets, especially Smad7 and Spry1[20, 21]. Because SSc has the unique feature of autoimmunity and vasculopathy characteristically precede fibrosis, and is distinguished from other connective tissue diseases and organ-specific fibrosing disorders[3, 23]. So we investigated the function of miR-21 in SSc fibrosis. Our studies demonstrated the same miR-21 expression and pathogenesis in SSc fibrosis. Targeting miR-21 may represent a potential therapeutic strategy. Up-regulation of miR-21 contributes to SSc fibrosis, and strategies to maintain normal miR-21 expression may be beneficial in SSc.

In bleomycin-treated mice, miR-21 expression was increased and decreased in bortezomib-treated group with amelioration of skin fibrosis. Changes in Smad7 expression were consistent with miR-21 expression. We speculate that an excessive TGF-β activity is involved in bleomycin-induced skin fibrosis, and that Smad7 can be strongly and immediately induced by TGF-β. When fibrosis was controlled by bortezomib, the TGF-β activity subsided, the Smad7 expression was decreased. TGF-β is an important mediator of fibrotic diseases, it caused miR-21 upregulation during fibrosis in our data, but it may not be the only factor, other pro-fibrotic growth factors, such as EGF signaling, bFGF, also regulate miR-21 expression[24, 25]. The relationship between miR-21 and Smad7 maybe more complicated in vivo [26]. Thus, dysregulation of miR-21 could arise from aberrations in multiple critical signaling events involved in SSc fibrosis.

MiR inhibitors might represent an effective means of decreasing miR-21 expression and thereby diminishing fibrosis, but the major drawback for miR inhibitors used in vivo studies is the high cost. Therefore, the use of much less expensive small molecules that are already available but not known to target miRNAs, or of new small molecules derived from known ones, will represent an alternative approach for the scientific community[27]. Pharmacological inhibitors to prevent the up-regulation of miR-21 expression during fibrosis may represent an intriguing therapeutic approach. Bortezomib a small molecular, proteasome inhibitor approved for clinical use in the treatment of multiple myeloma and mantle cell lymphoma [28] . In animal models, bortezomib can prevent fibrosis in the liver, kidney, lung, skin, and heart, and acts to inhibit a common pathway involved in organ fibrosis [17, 29–32]. Ballabio et al. reported that in addition to targeting the proteasome directly, bortezomib alters the expression of specific miRNAs that target downstream components of the protein ubiquitination pathway presumably acting in a synergistic fashion[33]. Our study suggests that bortezomib can decrease the expression of miR-21, how miR-21 was deregulated by bortezomib remains unclear, but nevertheless represents a novel therapeutic alternative in SSc.

Our studies identified miR-21 exerts its profibrogenic activity of regulating SSc fibrosis via Smad7, and suggest a novel approach using small molecules targeting miRNAs in treating SSc fibrosis. As miRNAs circulating as stable lipid-enclosed exosomes can be measured in blood [34, 35], they represent potential biomarkers of fibrosis. Kawashita et al. reported serum levels of miR-29a may be useful for the differentiation of SSc from SSc spectrum disorder [36]. Whether serum levels of miR-21 are also associated with the degree of fibrosis is an issue that will require more and larger studies.

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Conflict of interest statement The authors declare that there are no conflicts of interest.

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